AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	NV facility NV_way_775412073 -- station KRNO, America/Los_Angeles
Committed pair	OSM 775412069 -> 775412067 OSM way 775412069 / OSM way 775412067
Facade gap	112.9 m between the two halls
Weather record	43,687 real hourly records from KRNO
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3104 C at 250 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-03-23 (knife_edge)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 13 of 24 hours with 2 mode changes. The reactive incumbent took 24 hours -- 11 more -- but declared 0 unsafe hours against the agent's 0. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	13 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	24 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s) 10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	-0.719	18.0	1.270	4.383
01	FREE	yes	-	-0.408	18.0	0.741	4.336
02	FREE	yes	-	-0.159	18.0	0.720	3.910
03	FREE	yes	-	0.392	18.0	0.720	4.085
04	FREE	yes	-	0.392	18.0	0.720	3.745
05	FREE	yes	-	0.392	18.0	0.720	3.294
06	FREE	yes	-	0.942	18.0	1.270	2.748
07	FREE	yes	-	2.062	18.0	1.830	2.329

08	FREE	yes	-	4.841	18.0	2.940	2.285
09	FREE	yes	-	8.482	18.0	5.119	3.091
10	FREE	yes	-	12.611	18.0	7.380	4.279
11	FREE	yes	-	15.507	18.0	8.595	4.808
12	FREE	yes	-	17.618	18.0	8.049	5.193
13	mechanical	no	dry-bulb	18.018	18.0	9.159	5.446
14	mechanical	yes	-	16.837	18.0	10.265	4.812
15	mechanical	yes	-	16.831	18.0	9.134	4.999
16	mechanical	yes	switch budget	15.225	18.0	9.079	4.067
17	mechanical	yes	switch budget	13.216	18.0	7.313	3.697
18	mechanical	yes	switch budget	9.672	18.0	5.212	3.559
19	mechanical	yes	switch budget	8.241	18.0	4.662	3.143
20	mechanical	yes	switch budget	7.365	18.0	4.176	3.107
21	mechanical	yes	switch budget	6.092	18.0	3.081	3.600
22	mechanical	yes	switch budget	6.332	18.0	3.387	4.318
23	mechanical	yes	switch budget	5.406	18.0	4.168	4.588

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.719 C, which is 18.719 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.383 C, and the margin is measured, not chosen: 4.151 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.408 C, which is 18.408 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.336 C, and the margin is measured, not chosen: 4.183 C of group-conditional forecast error for this hour of day, plus 0.1523 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.159 C, which is 18.159 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.910 C, and the margin is measured, not chosen: 3.679 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.391 C, which is 17.609 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.085 C, and the margin is measured, not chosen: 3.853 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.391 C, which is 17.609 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.745 C, and the margin is measured, not chosen: 3.513 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.391 C, which is 17.609 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.294 C, and the margin is measured, not chosen: 3.062 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.941 C, which is 17.059 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.748 C, and the margin is measured, not chosen: 2.516 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 2.061 C, which is 15.939 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.328 C, and the margin is measured, not chosen: 2.097 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.841 C, which is 13.159 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.285 C, and the margin is measured, not chosen: 2.053 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.482 C, which is 9.518 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.091 C, and the margin is measured, not chosen: 2.950 C of group-conditional forecast error for this hour of day, plus 0.1409 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.611 C, which is 5.389 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.279 C, and the margin is measured, not chosen: 4.047 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.507 C, which is 2.493 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.808 C, and the margin is measured, not chosen: 4.576 C of group-conditional forecast error for this hour of day, plus 0.2319 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.618 C, which is 0.382 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.193 C, and the margin is measured, not chosen: 4.985 C of group-conditional forecast error for this hour of day, plus 0.2086 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.018 C against a 18.0 C limit, so it fails by 0.018 C. It would take a limit 0.018 C higher, or a bound 0.018 C tighter, to change this hour.

14:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.837 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.831 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.225 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.216 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.672 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.241 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.365 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.092 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.332 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.406 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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